EXP 5: COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS :

Aim: To test and compare how different pattern models respond to various prompts (broad or unstructured) versus basic prompts (clearer and more refined) across multiple scenarios. Analyze the quality, accuracy, and depth of the generated responses.

Explanation

In this experiment, the responses generated by an AI model are analyzed using two different types of prompts: naïve prompts and basic prompts. A naïve prompt is a simple and unstructured instruction that does not provide enough context or detail for the AI model. As a result, the response generated may be general, short, or lacking depth. On the other hand, a basic prompt is clear, structured, and provides specific instructions or context. This helps the AI model understand the task better and generate more accurate and detailed responses.

To perform the experiment, multiple test scenarios are selected. These scenarios include creative story generation, answering factual questions, summarizing a concept, and providing advice or recommendations. For each scenario, two prompts are created: one naïve prompt and one basic prompt. Both prompts are designed to perform the same task but differ in the level of detail and structure.

The naïve prompt is first given to the AI model, and the generated response is recorded. After that, the corresponding basic prompt is provided, and its response is also recorded. This process is repeated for all selected scenarios to obtain a complete set of results.

The responses obtained from both types of prompts are then compared based on quality, accuracy, and depth. Quality refers to how well the response is written and organized. Accuracy measures whether the information provided is correct and relevant to the question. Depth refers to how detailed and comprehensive the response is.

Through this comparison, the experiment helps to understand how prompt clarity and structure influence the performance of AI systems. It also demonstrates that well-structured prompts usually produce more meaningful and useful responses compared to vague prompts. The results provide insights into how users can design effective prompts to obtain better outputs when using AI tools like ChatGPT.

Define the Two Prompt Types

1. Naïve Prompt

A naïve prompt is a simple, broad, or unstructured instruction given to an AI model without providing sufficient details or context. It usually asks a general question and does not specify the format, length, or focus of the response. Because of this, the AI may generate answers that are basic, less detailed, or not fully focused on the user's needs.

Example:

"Explain Artificial Intelligence."

2. Basic Prompt

A basic prompt is a clear, structured, and detailed instruction given to an AI model. It provides specific context, requirements, or constraints such as the topic focus, format, or length of the response. This helps the AI understand the task better and produce responses that are more accurate, detailed, and well-organized.

Example:

"Explain Artificial Intelligence, including its definition, main types, and two real-world applications in about 100 words."

Basic Prompt

A basic prompt is a clear, detailed, and structured instruction given to an AI model. It includes specific context, format, or constraints that guide the model to produce more accurate and meaningful responses.

Example Basic Prompt:

"Write a 150-word story about a young warrior protecting a magical forest. Include a setting, a challenge the warrior faces, and how the warrior overcomes it."

Prepare Multiple Test Scenarios

To analyze the effectiveness of prompting patterns, the following scenarios are selected:

Creative Story Generation

Factual Question Answering

Concept Summarization

Advice or Recommendation

For each scenario, both naïve prompts and basic prompts are created.

Test Scenarios with Prompts

Scenario	Naïve Prompt	Basic Prompt
Creative Story	Write a story.	Write a 150-word story about a young warrior protecting a magical forest. Include the setting, conflict, and resolution.
Factual Question	What is climate change?	Explain climate change, its main causes, and two effects on the environment in about 100 words.
Summarization	Summarize Artificial Intelligence.	Summarize Artificial Intelligence including its definition, key features, and two applications in 80 words.
Advice	Give study tips.	Provide 5 effective study tips for engineering students preparing for semester exams.

Run Experiments with ChatGPT

The experiment is conducted using the following steps:

Input the naïve prompt into ChatGPT.

Record the generated response.

Input the corresponding basic prompt.

Record the generated response again.

Repeat the process for all selected scenarios.

This helps collect a complete set of responses for comparison.

Evaluation of Responses

The responses are evaluated using three criteria:

1. Quality

Quality refers to how well the response is written and organized.

2. Accuracy

Accuracy checks whether the information provided is correct and relevant.

3. Depth

Depth measures how detailed and comprehensive the response is.

Comparison Table

Scenario	Prompt Type	Quality	Accuracy	Depth
Creative Story	Naïve Prompt	Moderate	Good	Limited
Creative Story	Basic Prompt	High	Good	Detailed
Factual Question	Naïve Prompt	Moderate	Good	Limited
Factual Question	Basic Prompt	High	Very Accurate	Detailed
Summarization	Naïve Prompt	Average	Good	Short
Summarization	Basic Prompt	High	Accurate	Clear and Structured
Advice	Naïve Prompt	Moderate	Good	Basic Tips
Advice	Basic Prompt	High	Accurate	Detailed Suggestions

Analysis

The results show that prompt clarity significantly affects AI performance.

Basic prompts generally produce higher quality responses.

They provide better accuracy and more detailed explanations.

Naïve prompts can still work for very simple questions, but they often lack depth and structure.

For tasks like storytelling, summarization, and recommendations, structured prompts give much better results.

Summary of Findings

Clear prompts lead to better AI responses.

Providing context helps the model understand the task more effectively.

Structured prompts improve quality, accuracy, and depth of answers.

Users should design specific and detailed prompts to obtain optimal results when using AI tools like ChatGPT.

Result:

Basic promps are quick but lack but refined prompts produce superior result.As refined prompt is best for getting better fit result and approximation